
When Robustness Metrics Mislead: Measuring Cue Effects on LLM Judge Correctness

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Abstract

1 Robustness of a large language model (LLM) judge is often summarised by the
2 fraction of verdicts that change following an injected cue. This scalar flip rate con-
3 flates corrections and corruptions despite their opposite effects on accuracy. Where
4 ground truth is available, the direction of each flip is known and the net accuracy
5 change $(R - C)/N$ can be calculated directly. A common alternative compares
6 the conditional repair rate among initially incorrect questions with the conditional
7 corruption rate among initially correct questions. The two rates therefore have
8 different denominators, and their difference is sensitive to the composition of the
9 corresponding baseline subsets. Across 499 questions and five judges, $(R - C)/N$
10 is negative for every judge, from -0.204 to -0.411 , while the conditional asymme-
11 try is positive for every judge, from $+0.058$ to $+0.437$: the two statistics disagree
12 in sign on all five, with no binning involved. We address this discrepancy using
13 a paired measure of the change in probability assigned to the correct answer
14 under both cue directions for every question, checked against a simulated judge
15 with a known constant effect. Fourteen of fifteen judge $\times$ bin cells are negative,
16 including every middle and high confidence cell. Under balanced cue exposure,
17 the probability of a correct verdict in the high confidence tercile falls by 0.20 to
18 0.28 . Prompt protocol changes the rate asymmetry by 0.165 and 0.069 on two
19 judges, with each spread exceeding at least one difference between judges. For
20 both judges evaluated under all four protocols using the paired measure, the sign
21 remains negative throughout.

22 1 Introduction

23 Language models are increasingly used to evaluate other models' outputs [Zheng et al., 2023, Wang
24 et al., 2024], despite evidence that such evaluations respond to cues unrelated to answer quality:
25 position bias [Wang et al., 2024, Shi et al., 2025], self-preference [Panickssery et al., 2024], verbosity
26 [Saito et al., 2023], and preference for answers matching a user's view [Sharma et al., 2024, Perez
27 et al., 2023]. Robustness is commonly summarised by the fraction of verdicts changing when an
28 injected cue is swapped.¹ Without ground truth, that rate records whether a verdict changed, not
29 whether the change improved or reduced accuracy. The correction is known: Lee et al. [2025] pair a
30 direction free switch rate with an accuracy change and argue a robust judge should show zero accuracy
31 change, and De Marez et al. [2026] decompose flip rates into a truth margin and a manipulation
32 sensitivity across 56 models. Our contribution is instead that the natural refinement, normalising each
33 direction by its own opportunities, is composition sensitive enough to reverse sign, with a paired
34 measure that avoids the comparison.

¹We use *flip rate* for that paired quantity. Competing names for closely related quantities are set out in Appendix F.

Judge	within	comp.	ret.	aggregate	flip
Llama-3.1-8B	-0.160	+0.068	+0.150	+0.058	yes
Qwen2.5-7B	+0.212	-0.006	+0.107	+0.314	no
Qwen2.5-14B	+0.267	+0.000	+0.170	+0.437	no
Mistral-7B	+0.044	+0.036	+0.000	+0.081	no
Phi-4-mini	+0.035	+0.015	+0.108	+0.158	no

Table 1: Decomposition of the ground truth rate asymmetry from Equation 1. The three terms sum to the aggregate by construction, each rounded independently, which accounts for a final decimal mismatch in two rows. Across four judges the composition and retention terms together account for 32 to 78% of the aggregate; for Llama-3.1-8B they exceed it and reverse its sign, and retention is zero only for Mistral-7B, where no quintile is dropped. Bootstrap intervals for every term are in Appendix H; Llama-3.1-8B’s within term excludes zero while its aggregate does not.

Ground truth permits each changed verdict to be classified by its effect on correctness rather than counted without direction. A change from a correct baseline verdict to the wrong option is a corruption, whereas a change from an incorrect baseline verdict to the correct option is a repair. Zhao et al. [2026], for example, report that pressure induced flips are predominantly net corrupting, consistent with the direction of our paired results. This distinction yields two summaries with a common denominator: net accuracy change $(R - C)/N$ and total instability $(R + C)/N$, each evaluated over the full set of N questions.

The central measurement problem arises when corruption and repair are normalised separately, since the repair rate is conditional on questions answered incorrectly at baseline and the corruption rate on questions answered correctly. Baseline correctness therefore determines which denominator each question enters, and if confidence is distributed differently across the two subsets the resulting asymmetry may reverse sign. We study this through controlled changes to the evaluator input, following the contrast set principle [Gardner et al., 2020] and the “right for the wrong reasons” problem [McCoy et al., 2019]. As a measurement problem this concerns construct validity [Cronbach and Meehl, 1955], of which annotation artefacts are a related example [Gururangan et al., 2018].

2 Setup

Questions. The Balanced COPA development set contains 500 mirrored pairs [Kavumba et al., 2019]. We retain one question from each pair, thereby avoiding repeated presentation of the same alternatives. In the version used for our experiments, the pair with ids 381 and 1381 is identical in premise and in both alternatives while carrying opposite labels; its exclusion leaves 499 two-choice questions with known correct answers.

Judges and cues. We evaluate Llama-3.1-8B, Qwen2.5-7B, Qwen2.5-14B, Mistral-7B and Phi-4-mini, all instruction tuned. Each cue is a sentence inserted before the options. An *authority* cue attributes one option to a named expert, a *sycophancy* cue states the user’s belief, and a *length* cue appends semantically irrelevant text. The cues are question specific rather than instances of a single template reused verbatim. On the two judges for which we evaluated both forms, template based cues change the asymmetry by -0.105 and $+0.036$; neither interval excludes zero.

Verdicts come from the next token log probabilities of the two option labels, renormalised over those labels alone: $p_{\text{correct}} = e^{\ell_c} / (e^{\ell_c} + e^{\ell_w})$. Every label is a single token in the output context for every judge and protocol, removing the need for generation or a parser. Option order follows the source dataset here and is counterbalanced in Section 5. Uncertainty comes from percentile bootstraps over *questions*, 5000 resamples, a question’s baseline and both cue directions held together; the fifteen intervals in Figure 1 are pointwise, and multi-cell claims use Holm’s correction [Holm, 1979]. Questions are grouped within judge by $|\ell_0|$: terciles for the paired analysis, quintiles for the decomposition.

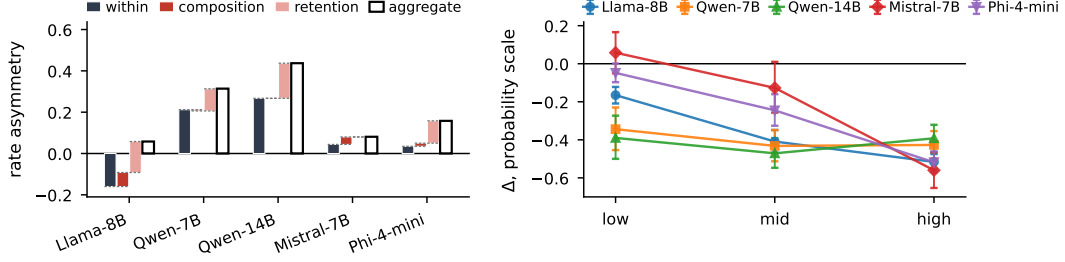


Figure 1: *Left*: Equation 1 as a waterfall. Each judge’s within, composition and retention terms are drawn in sequence, each starting where the previous ended, and the open bar is the aggregate they sum to. On Llama-3.1-8B the within term is negative while the two artefact terms carry the aggregate above zero. *Right*: Δ against the judge’s own baseline confidence, by bin, with 95% bootstrap intervals over questions. Twelve of fifteen cells exclude zero; the exceptions are Mistral-7B at low and middle confidence and Phi-4-mini at low.

70 3 Conditional repair and corruption rates are composition sensitive

71 *All results in this section use the authority cue.* Comparisons of this form appear in the sycophancy
 72 literature, separating a flip that corrects an initial error from one that abandons an initial truth [Chang,
 73 2026], or a progressive from a regressive rebuttal response [Fanous et al., 2025]. The disagreement
 74 needs no binning: on the common denominator $(R - C)/N$ is negative for every judge, from -0.204
 75 to -0.411 , while $q - \gamma$ is positive for every judge, from $+0.058$ to $+0.437$, with baseline subsets
 76 ranging from 30 to 469 questions (Appendix B). The decomposition below is a mechanism for that
 77 disagreement rather than its primary evidence; Appendix G sets out why the conditional form is used
 78 at all.

79 Let α denote the aggregate asymmetry, repair minus corruption. For bin k of $|\ell_0|$, let c_k and r_k be the
 80 proportions of the baseline correct and baseline incorrect subsets in that bin, with rates γ_k and q_k and
 81 within-bin asymmetry $\alpha_k = q_k - \gamma_k$. The aggregate is $\alpha = \sum_k (r_k q_k - c_k \gamma_k)$ and the dependence
 82 on composition enters through differences between r_k and c_k . Questions are divided into quintiles,
 83 a bin retained only when both subsets hold ten; S is the retained set and α_S the asymmetry on it.
 84 Standardising both subsets to the pooled distribution w_k over S yields

$$\alpha = \underbrace{\sum_{k \in S} w_k \alpha_k}_{\text{within}} + \underbrace{\sum_{k \in S} (r_k - w_k) q_k - (c_k - w_k) \gamma_k}_{\text{composition}} + \underbrace{\alpha - \alpha_S}_{\text{retention}}, \quad (1)$$

85 an identity. Composition is zero when the two baseline subsets have the same distribution across
 86 retained bins, retention when every bin is retained. Both characterise the denominators rather than a
 87 response property of the judge.

88 Table 1 and Figure 1 trace the disagreement to composition and retention. For Llama-3.1-8B the
 89 aggregate is $+0.058$ while the three retained quintiles are negative at -0.163 , -0.163 and -0.152 :
 90 a Simpson type reversal [Simpson, 1951, Pearl, 2014]. Of the $+0.218$ gap, retention contributes
 91 $+0.150$ and composition $+0.068$; neither reverses the sign alone, since within plus composition is
 92 -0.092 and within plus retention -0.009 . Retention is larger here and on three of the other four
 93 judges. The two dropped quintiles hold 195 questions in the corruption denominator against 5 in the
 94 repair.

95 Aggregate asymmetry correlates with baseline accuracy at $r = +0.81$, although with $n = 5$ this is not
 96 significant ($p = 0.10$; Spearman $\rho = 0.70$, $p = 0.19$) and the bootstrap interval spans $[-0.30, +1.00]$.
 97 The argument rests on the within judge comparison. Across the 28 settings obtained by crossing bin
 98 counts from three to ten with thresholds from five to thirty questions, the Llama-3.1-8B within term
 99 remains between -0.114 and -0.209 , while the aggregate remains $+0.058$ by construction.

100 4 A paired measure of cue induced correctness

101 Comparison of two separately normalised rates can change the sign of the apparent effect. We adopt a
 102 paired quantity for which every question contributes to both cue directions. Let p_0 be the probability

assigned to the correct option without a cue, and p^+ , p^- the same probability when the cue points to the correct and incorrect option. We define

$$\Delta = (p^+ - p_0) + (p^- - p_0) = p^+ + p^- - 2p_0, \quad (2)$$

with both directions measured on every question. Each question is compared with its own baseline, holding confidence and question identity fixed. The composition problem of Section 3 is absent by construction rather than addressed by matching.

Interpretation of Δ . The estimand is net correctness: equal and opposite cue effects yield $\Delta = 0$ whatever their magnitude, directional sensitivity being the separate quantity $(p^+ - p^-)/2$. Since Δ sums the two shifts, balanced cue exposure costs $\Delta/2$; we report that and plot Δ .

Figure 1 reports Δ by confidence bin. Fourteen of the fifteen cells are negative, including every middle and high confidence cell. At high confidence Δ ranges from -0.39 to -0.56 , a 0.20 to 0.28 fall in the probability of a correct verdict under balanced exposure. Twelve of the fifteen survive Holm at $\alpha = 0.05$; the three that do not are Mistral-7B at low and middle confidence and Phi-4-mini at low, exactly those whose intervals contain zero. Every surviving cell is negative, which the normalised rate asymmetry is not; three judges sit close to outright cue following (Appendix E). On Llama-3.1-8B and the same questions, the sycophancy and length cues give $\Delta = -0.355$ and -0.119 against -0.364 for authority. These runs are not pooled, although the shared sign places the effect beyond the authority framing.

Choice of scale. Pairing removes the composition problem, although the value remains scale dependent and neither scale is neutral. We use probability because the quantity of practical interest is the change in the chance of a correct verdict; Appendix I gives the two bounding simulations.

The confidence gradient does not identify a confidence dependent effect. For Llama-3.1-8B, Mistral-7B and Phi-4-mini, $|\Delta|$ increases monotonically across the confidence terciles, from 0.165 to 0.516 , 0.058 to 0.560 and 0.048 to 0.518 . Δ is curved in p_0 , and two confidence independent nulls fitted on the lowest tercile alone bracket the observed gradient, one over predicting the highest tercile and the other under predicting it (Appendix I). The gradient is therefore not evidence of greater susceptibility where confidence is higher.

5 Sensitivity of the rate asymmetry to prompt format

The paired construction removes the composition dependence, although a robustness estimate can remain sensitive to how the prompt is presented. We repeat the measurement under four prompt protocols for two judges, differing in more than wording: one omits the instruction, another passes the prompt through the chat template.

Sensitivity to prompt formatting and example order is documented [Sclar et al., 2024, Lu et al., 2022, Zhao et al., 2021]; here it changes the robustness estimate itself, questions and readout fixed, by an amount differing across judges (Table 3). Llama’s spread of 0.165 exceeds five of the ten pairwise differences in Table 1, and Qwen’s 0.069 exceeds the 0.022 between Llama and Mistral. A comparison under a single protocol may therefore conflate a difference between judges with one induced by the protocol.

Sensitivity to option order. Counterbalancing the two option positions changes the rate asymmetry by as much as 0.155 for Qwen2.5-14B, exceeding several differences between judges, while the sign of the paired effect is stable. After averaging over option order, Δ remains negative for every judge, at -0.365 , -0.408 , -0.457 , -0.250 and -0.282 , with every interval excluding zero.

6 Conclusion

A flip rate cannot say whether a judge was corrected or corrupted; ground truth can. Normalising each direction by its own opportunities then reintroduces a confound strong enough to invert the sign. The paired measure avoids that comparison and reads a continuous probability rather than a thresholded verdict, discarding less signal than a count of flips. Limitations: Appendix C.

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A Code and reproducibility

Code for reproducing the experiments and analyses is available at

<https://anonymous.4open.science/r/NeurIPS2026-JUDGE-361E/>

B Corruption and repair counts

Section 3 defines corruption and repair relative to the known correct answer. This appendix reports the counts underlying those rates for $N = 499$ questions. A question contributes to the corruption rate denominator when the baseline verdict is correct and to the repair rate denominator when the baseline verdict is wrong. These two baseline subsets partition the dataset, and their sizes therefore depend on the judge’s baseline accuracy. We write C for questions from the baseline correct subset whose verdict moves to the wrong option under the cue, and R for questions from the baseline incorrect subset whose verdict moves to the correct option.

Judge	baseline subset sizes		counts		rates		common denominator	
	correct	incorrect	C	R	γ	q	$(R - C)/N$	$(R + C)/N$
Llama-3.1-8B	407	92	224	56	0.550	0.609	−0.337	0.561
Qwen2.5-7B	453	46	242	39	0.534	0.848	−0.407	0.563
Qwen2.5-14B	469	30	233	28	0.497	0.933	−0.411	0.523
Mistral-7B	315	184	281	179	0.892	0.973	−0.204	0.922
Phi-4-mini	377	122	271	107	0.719	0.877	−0.329	0.758

Table 2: Corruption and repair counted separately on the common denominator.

The common denominator gives the same directional conclusion for all five judges, $(R - C)/N$ between −0.204 and −0.411, while $q - \gamma$ in Table 1 is positive for all five because it answers a different question: its two rates are conditional on different baseline states, whose subsets range in size from 30 to 469, whereas $(R - C)/N$ compares both directions on the same N questions.

C Limitations

Δ needs three forward passes and log probabilities, which excludes judges accessible only through an API, and its assumptions require validation of their own. The experiments cover one dataset, one readout and five judges from 4B to 14B. Bins are defined on each judge’s own margin, so Δ magnitudes are not compared across judges.

The format is narrower than the motivating literature: two-choice questions with an inserted sentence rather than candidate response pairs, and biases defined over pairs are not exercised. The composition argument does not depend on the format, although transfer to pairwise judging is untested.

D Prompt protocols

Protocol	Llama asym.	Qwen asym.	Qwen Δ
plain, with instruction	+0.058	+0.314	−0.401
no instruction	+0.134	+0.303	−0.509
chat template	+0.121	+0.299	−0.462
question first	+0.223	+0.244	−0.494
spread	0.165	0.069	0.108

Table 3: Rate asymmetry and paired effect under four prompt protocols.

Stability of the sign under option counterbalancing does not imply invariance of the paired measure in magnitude. Across the four protocols Δ spans 0.108 on Qwen2.5-7B and 0.118 on Llama-3.1-8B,

negative throughout on both, which supports stability of the sign over these protocols rather than a general claim of format invariance.

Protocol sensitivity is not confined to the conditional rates. Measured against the pairwise differences between judges in the same quantity, the protocol spread in $(R - C)/N$ exceeds ten of ten on Llama-3.1-8B and six of ten on Qwen2.5-7B, where the rate asymmetry exceeds five and one. The common denominator removes the composition dependence of Section 3, not the dependence on how the judge is asked: a protocol change alters what the judge does, and both statistics record that. What separates them is that the asymmetry also moves with the composition of its two denominators on data that is otherwise fixed.

E Saturation

Three judges sit close to outright cue following, Mistral-7B, Qwen2.5-14B and Phi-4-mini, with repair rates of 0.973, 0.933 and 0.877. The two least saturated, Llama-3.1-8B at 0.609 and Qwen2.5-7B at 0.848, leave all six of their cells negative, so the sign of the paired effect does not depend on the saturated cases.

F Terminology

Three names are in circulation for closely related quantities. Marioriyad et al. [2025] define a paired Verdict Shift Rate in their methods, then use the same term for a signed difference in selection rates; we use *flip rate* for the paired quantity alone. Lee et al. [2025] use Verdict Switch Rate for the sum of changes in both directions, which is our instability $(R + C)/N$, and pair it with an accuracy change corresponding to our $(R - C)/N$.

G Why the conditional form is used

The conditional form has a defensible motivation: it makes the rates invariant to how many questions the judge got right, which is what a reader comparing judges of different accuracy wants. Our objection is not that $q - \gamma$ is ill defined, rather that the invariance is bought by conditioning on subsets of differing composition, and the result is then read as a statement about robustness when it is partly one about accuracy.

H Intervals for the decomposition

Percentile bootstraps over questions, 5000 resamples, with the quintile edges and the retention rule recomputed on each resample, and the interval therefore covers the binning choice rather than conditioning on it.

Judge	within	composition	retention	aggregate
Llama-3.1-8B	$[-0.296, -0.033]$	$[+0.009, +0.141]$	$[+0.093, +0.224]$	$[-0.053, +0.169]$
Qwen2.5-7B	$[+0.008, +0.322]$	$[-0.041, +0.043]$	$[+0.027, +0.266]$	$[+0.194, +0.421]$
Qwen2.5-14B	$[+0.126, +0.427]$	$[+0.000, +0.000]$	$[+0.018, +0.284]$	$[+0.328, +0.526]$
Mistral-7B	$[-0.016, +0.085]$	$[+0.007, +0.061]$	$[+0.000, +0.060]$	$[+0.040, +0.121]$
Phi-4-mini	$[-0.060, +0.120]$	$[-0.001, +0.100]$	$[+0.035, +0.159]$	$[+0.083, +0.231]$

Table 4: Bootstrap intervals for the three terms of Equation 1 and for the aggregate.

Three within terms exclude zero, those of Llama-3.1-8B, Qwen2.5-7B and Qwen2.5-14B. Llama-3.1-8B’s aggregate does not; its sign reversal is between a within term that is significantly negative and an aggregate that is not distinguishable from zero.

Qwen2.5-14B’s composition interval is degenerate at zero because its baseline incorrect subset holds only 30 questions, so the retention rule leaves a single bin in every resample. The within term is then that bin’s α_k and composition vanishes by construction rather than by measurement.

I Scale, and the nulls the confidence gradient lies between

Scale. On the log odds scale a simulated judge with a constant probability-scale effect of -0.060 yields -0.277 , -0.684 and -5.208 across confidence terciles: a constant effect on one scale appears as a steep gradient on the other. The probability scale has its own curvature effect: a symmetric logit response can produce a negative Δ , and ± 3 logits at $p_0 = 0.9$ give $\Delta = -0.496$. Neither observation rules out curvature as part of the measured pattern; we report on the probability scale because the deployment question concerns the change in the chance of a correct verdict, and analyses directed at mechanism may prefer log odds.

Two nulls for the gradient. $\Delta = p^+ + p^- - 2p_0$ is curved in p_0 , and a judge with no confidence dependent behaviour can still show a gradient. We fit two one-parameter nulls on the *lowest* tercile alone and extrapolate, which neither can do by absorbing a genuine confidence dependence. Partial cue following, $\Delta = \lambda(1 - 2p_0)$, gives λ between 0.70 and 0.95 and *over* predicts the highest tercile on all five judges, by $+0.124$ to $+0.320$ with every interval excluding zero. A constant shift in log odds, fitted the same way, *under* predicts it by -0.273 to -0.422 . The observed gradient lies between the two, so neither null is ruled out and the gradient does not distinguish a confidence dependent response from a confidence independent one. We therefore report it as description rather than as a property of the judges.

J Retained and dropped bins for Llama-3.1-8B

The sign reversal in Table 1 depends in part on which confidence bins satisfy the retention rule. A bin is retained only when each baseline subset contains at least ten questions. For Llama-3.1-8B, the first three quintiles meet this condition and the final two do not:

Bin	baseline correct	baseline incorrect	α_k	status
1	49	49	-0.163	retained
2	79	23	-0.163	retained
3	84	15	-0.152	retained
4	96	4	–	dropped
5	99	1	–	dropped

Table 5: Retained and dropped bins for Llama-3.1-8B.

The retained bins receive weights 0.328, 0.341 and 0.331, producing a within term of -0.160 . The final two bins are excluded because the baseline incorrect subset contains only 4 and 1 questions in those bins; together the two bins contain 195 baseline correct questions and 5 baseline incorrect questions. Their exclusion therefore changes which portions of the two baseline subsets contribute to the within term. The aggregate over all questions exceeds the within term by 0.218 even though α_k is negative in every retained bin. In the decomposition of Equation 1, retention accounts for 0.150 of this difference and composition for the remaining 0.068.

AI declaration. Generative AI tools were used for text editing and code assistance.